

PRODUCT REQUIREMENTS DOCUMENT

AI-Powered Credit Card Recommendation Engine

1. Product Overview

The AI-Powered Credit Card Recommendation Engine transforms the traditional credit card discovery experience by replacing static, form-based journeys with an intelligent, behaviorally-driven recommendation system built natively on the digital banking platform.

The engine leverages a user's consented financial footprint SMS transaction data, bureau information, spending patterns, and stated preferences to understand behaviour, assess eligibility in real time, and match users with the most relevant credit card based on spending behavior, eligibility, and financial goals. The result is a low-friction, high-trust product journey that reduces user effort and measurably improves application conversion rates.

2. Problem Statement

The current credit card selection process is fragmented, high-effort, and poorly personalised for the Indian digital consumer.

Pain Points Observed

- **Questionnaire fatigue:** Existing journeys rely on long, manual forms asking users to self-report income, spending, and goals resulting in 50–60% drop-off before completion.
- **Irrelevant recommendations:** Generic eligibility filters surface cards that do not reflect actual user behaviour, eroding trust in the platform.
- **Opaque logic:** Users receive recommendations with no explanation, reducing confidence and willingness to apply.
- **High application failure rates:** Poor eligibility estimation leads users to apply for cards they are unlikely to receive, damaging user experience and bureau scores.
- **Missed monetisation:** Misaligned recommendations result in lower card issuance rates and reduced revenue per user for the platform.

Opportunity

By intelligently ingesting consented financial signals, the platform can reduce manual input to near-zero, produce high-confidence recommendations within 90 seconds, and deliver transparent reasoning that converts browsers into applicants.

3. Product Objectives

- Minimise onboarding friction through intelligent, consent-based data collection.
- Deliver personalised card recommendations grounded in real financial behaviour not self-reported proxies.
- Surface the top 3 most relevant credit card recommendations with match scores, approval probability, and estimated annual benefit.
- Build user trust through fully explainable recommendation reasoning.
- Increase recommendation-to-application conversion rate from ~12% (industry baseline) to 25%+.
- Reduce average time-to-recommendation to under 90 seconds.
- Improve card issuance success rate by surfacing only eligible options.

4. Scope & Constraints

4.1 V1 in Scope

- Mobile-first recommendation journey on the digital banking application.
- Consent-based SMS parsing for financial signal extraction.
- Soft-pull bureau integration for credit health signals.
- Rule-based persona classification engine (ML upgrade in V2).
- Weighted scoring recommendation model covering the full supported credit card portfolio.
- Explainability layer surfacing top 3 recommendation reasons per card.
- Pre-filled application handoff to issuing bank application system.
- Graceful fallback flows for users with denied consent or no credit history.

4.2 Out of Scope for V1

- Conversational / chatbot recommendation interface (V2 roadmap).

- Real-time transaction sync via Account Aggregator (V2).
- Credit card products outside the supported card portfolio.
- Hard bureau pull (only soft pull in V1 to protect user CIBIL score).
- ML-based persona clustering rule-based logic only in V1.
- Cross-sell into loans, insurance, or savings products.
- Web / desktop version of the journey.

4.3 Assumptions & Constraints

- SMS parsing occurs on-device; raw message content is never transmitted to platform servers; only extracted structured signals are sent.
- Bureau pull is a soft pull (no CIBIL score impact). Hard pull occurs only post-application within the Bank system.
- The supported credit card portfolio for V1 consists of 8–12 active cards with defined eligibility criteria.
- Users must be Indian residents aged 18+ with a valid PAN.
- Minimum viable signal for a recommendation: at least 3 months of parseable SMS data OR bureau data. If neither is available, fallback to explicit preference collection.
- Compliance with RBI guidelines on digital lending and the Digital Personal Data Protection (DPDP) Act 2023 is mandatory.
- Account Aggregator (AA) framework integration is targeted for V2 pending regulatory licensing.

5. Target Users & Personas

Personas are behavioural archetypes derived from financial signals, not rigid demographic buckets. Users are dynamically mapped to a primary and (optionally) a secondary persona. A user can match multiple personas this enables nuanced, blended recommendations.

Persona	Profile	Dominant Spending	Core Goals	Card Need
Young Salaried Professional	Age 22–30, early-career, regular monthly salary, high UPI usage	Online shopping, food delivery, OTT, frequent small transactions	Cashback, dining rewards, low annual fee	High-value daily cashback card
Frequent Traveller	Age 28–45, mid–high income, domestic and international travel	Airlines, hotels, international FX transactions	Air miles, lounge access, travel privileges	Premium travel rewards card

Practical Saver	Age 30–45, budget-conscious, household manager	Fuel, utility bills, grocery, essential household spends	Everyday savings, fuel cashback, utility rewards	Fuel & utility optimised card
Credit Beginner	First-time applicant, thin credit file, stable income	Limited credit history, primarily UPI / debit transactions	Credit score building, easy approval, low risk exposure	Entry-level card, higher approval likelihood
Premium Lifestyle	Age 30–50, high income, high discretionary spend	Luxury retail, fine dining, golf, premium concierge	Premium privileges, concierge, high credit limit	Super-premium lifestyle card

6. Product Principles

- **Consent First:** User financial data is accessed only after explicit, granular, and revocable consent. Consent can be partial the engine degrades gracefully.
- **Low Friction by Design:** Every screen has a single job. Manual data entry is minimised through smart inference from consented signals.
- **Explainability as Trust:** Every recommendation surfaces the top 3 reasons it was made, in plain language. No black-box outputs.
- **Behaviour Over Demographics:** Recommendations are driven by what users actually spend money on, not self-reported income brackets or age ranges.
- **Eligibility Transparency:** Approval probability is shown upfront. Users are never nudged toward cards they are unlikely to receive.
- **Privacy by Architecture:** SMS parsing is on-device. Only structured signals never raw messages leave the user's phone.

7. Card Catalogue (V1)

The recommendation engine will operate over the following credit card recommendations. Eligibility criteria and reward structures below are indicative; final values to be confirmed with the Bank product team prior to development.

Card Name	Best-Fit Persona	Min CIBIL	Min Income (Monthly)	Headline Benefit
-----------	------------------	-----------	----------------------	------------------

Cashback Plus Card	Credit Beginner	650+	₹15,000	1% cashback on all spends; zero annual fee Y1
Everyday Cashback Card	Young Salaried Professional	700+	₹20,000	5% cashback on online shopping & food delivery
Smart Rewards Card	Young Salaried Professional	720+	₹25,000	5% cashback via GPay; 4% on utility bills
Lifestyle Rewards Card	Young Salaried Professional	700+	₹20,000	15% off dining; OTT subscription benefits
Online Shopping Card	Online Shopper	700+	₹20,000	5% cashback on Flipkart; 4% on preferred merchants
Travel Rewards Card	Frequent Traveller	750+	₹50,000	Complimentary Vistara tickets; Club Vistara points
Premium Travel Card	Frequent Traveller	750+	₹50,000	5 EDGE Miles per ₹200; multiple lounge access
Elite Lifestyle Card	Premium Lifestyle	800+	₹1,50,000	Unlimited lounge; golf privileges; concierge
Utility Saver Card	Practical Saver	730+	₹40,000	Accelerated EDGE Rewards on all spends
Credit Builder Card	Credit Beginner	No History	₹12,000	Entry secured card; credit-building focus

8. Core Features

Feature 1: Smart Onboarding & Consent Framework

Collects the minimum viable user information needed to begin financial analysis, with explicit and granular consent at each data layer.

Consent Layers (in order of collection)

- **Mobile OTP Verification:** Confirms identity; no financial data collected at this step.
- **PAN Verification:** Lightweight KYC; used to match bureau records and pre-fill the application.
- **SMS Read Consent:** On-device only. User is shown exactly what categories of messages will be parsed (salary, EMI, utility, shopping). No raw SMS data ever leaves the device.

- **Bureau Soft Pull Consent:** Informs user this is a soft pull with zero CIBIL score impact. Bureau is fetched only after explicit acceptance.
- **Bank Statement Upload (Optional):** For users who want to enhance recommendation confidence or whose SMS data is sparse.

Consent Fallback Rules

- If SMS consent is denied → switch to explicit preference collection (lightweight 5-question flow).
- If bureau consent is denied → recommendations shown without eligibility probability scores; user is informed.
- If both are denied → minimum viable flow: 5 explicit questions + card catalogue filtering by stated income.

Feature 2: AI Financial Footprint Analyzer

Converts raw, consented financial data into structured behavioural attributes that feed the recommendation engine.

Data Sources & Extracted Signals

Data Source	Signals Extracted	Output Attribute
SMS Data (on-device)	Salary credits, EMI debits, utility payments, merchant spends, travel bookings, fuel transactions	Income stability, category spend %, EMI obligation ratio
Bureau Report (soft pull)	CIBIL score, active credit lines, repayment history, credit utilisation ratio, derogatory marks	Credit Health Score, eligibility band, utilisation flag
Transaction History	Merchant category, spend frequency, average ticket size, recurring patterns	Spend consistency score, high-frequency category tags
Explicit User Inputs	Preferred reward type, occupation, income range, stated spending priorities	Preference alignment vector for model weighting

Engineered Behavioural Features

- Travel Affinity Score (0–100)
- Cashback Affinity Score (0–100)
- Fuel Dependency Score (0–100)
- Dining & Lifestyle Score (0–100)
- Online Shopping Intensity Score (0–100)

- Credit Health Score (0–100)
- Salary Stability Score (0–100)
- EMI Burden Ratio (%)

Feature 3: Persona Classification Engine

Maps each user to a primary persona and (where applicable) a secondary persona using a rule-based decisioning tree in V1, upgradeable to ML clustering in V2.

Classification Logic (V1 Rule-Based)

- Travel Affinity Score > 65 AND salary > ₹50,000/month → Frequent Traveller
- Online Shopping Score > 60 AND age proxy < 30 → Young Salaried Professional
- Fuel Score > 55 AND utility payment regularity > 80% → Practical Saver
- No bureau file OR CIBIL < 680 → Credit Beginner (regardless of spend)
- Credit Health Score > 85 AND income > ₹1,20,000/month → Premium Lifestyle

Where a user scores high on two personas, both are retained. The recommendation engine uses the primary persona for card ranking and the secondary for tie-breaking and supplemental cards.

Feature 4: Dynamic Recommendation Engine

Ranks eligible cards using a weighted multi-factor scoring model. Only cards the user is likely eligible for are surfaced.

Recommendation Scoring Formula

$$\text{Recommendation Score} = (\text{Behavioural Fit} \times 0.40) + (\text{Eligibility Match} \times 0.30) + (\text{Reward Optimisation} \times 0.20) + (\text{Conversion Probability} \times 0.10)$$

Score Components Defined

Component	Definition
Behavioural Fit (40%)	How closely the card's reward structure aligns with the user's actual spending categories and affinity scores.

Eligibility Match (30%)	Probability of approval based on CIBIL score, income level, and existing credit obligations relative to the card's criteria.
Reward Optimisation (20%)	Estimated annual monetary value of rewards the user would earn based on their current spend volumes.
Conversion Probability (10%)	Platform-observed likelihood of this persona segment completing an application for this card type.

A hard eligibility gate is applied before scoring: cards where CIBIL < minimum threshold or income < minimum requirement are excluded entirely from the recommendation set, regardless of behavioural fit.

Feature 5: Explainability Layer

Every recommendation is accompanied by 3 plain-language reasons drawn directly from the user's financial signals. This is not an optional copy, it is dynamically generated per user.

Example Output Smart Rewards Card Card

- We recommend this card because:
1. 38% of your monthly spending is on online shopping. This card returns 5% on every online transaction.
 2. You pay utility bills regularly every month this card returns 4% on all utility payments.
 3. Your CIBIL score and income comfortably meet this card's eligibility criteria. Approval likelihood: High.

Feature 6: Eligibility Confidence Indicator

Displayed prominently on every recommendation card to set honest expectations and reduce wasted applications.

Indicator	Criteria	UI Treatment
High Approval Likelihood	CIBIL ≥ card minimum + 50pts; income ≥ 120% of minimum; utilisation < 30%	Green badge: 'You likely qualify'
Medium Approval Likelihood	CIBIL within 0–50pts of minimum; income meets minimum; some existing obligations	Amber badge: 'You may qualify'
Low Approval Likelihood	CIBIL below minimum OR income below minimum card shown only if best available match	Grey badge: 'Approval not guaranteed'

9. End-to-End User Journey

The journey is designed for mobile-first consumption. Target total time from landing to recommendation: under 90 seconds for users with sufficient SMS data.

Step 1: Entry Point

User arrives via: banking application home banner, targeted push notification, or paid acquisition campaign. Single full-screen CTA: 'Find Your Perfect Credit Card in 2 Minutes'.

Step 2: Value Proposition Screen

3-point trust message: Personalised for you | 100% Secure | No impact on CIBIL. Progress bar shown throughout journey. Single CTA: 'Get Started'.

Step 3: Data Consent & Verification

Mobile OTP → PAN entry → SMS consent screen (shows exactly what will be read) → Bureau consent screen (explains soft pull). Each consent step is a standalone screen with Accept / Skip options.

Edge Case: If SMS denied → skip to Step 5 (explicit preferences). If Bureau denied → proceed but suppress approval probability on results.

Step 4: Financial Analysis (Backend)

On-device SMS parsing extracts signals. Bureau API called in parallel. Structured signals sent to recommendation engine. User sees an animated 'Analysing your financial profile' loader with 3–4 rotating factoids. Target processing time: < 8 seconds.

Edge Case: If SMS parsing returns < 3 months of data → supplement with explicit preference questions in Step 5. If bureau pull fails → retry once, then proceed without bureau signals.

Step 5: Preference Collection (Contextual)

Shown only if (a) SMS consent was denied, or (b) insufficient SMS data. Max 5 questions: primary reward preference, monthly spending range, top 2 spending categories, employment type, existing credit cards. Skipped entirely if sufficient financial signals are already available.

Step 6: Recommendation Results Page

Top 1 card shown as 'Best Match' with full explanation. Cards 2 and 3 shown below as 'Also Recommended'. Each card shows: Match Score, Estimated Annual Benefit, Approval Likelihood badge, Top 3 recommendation reasons. Primary CTA: 'Apply Now'. Secondary CTA: 'See All 3 Cards'.

Edge Case: If no card meets the hard eligibility gate → show the Credit Builder Card secured card with copy: 'Start building your credit profile this card is designed for you.'

Step 7: Comparison View

Side-by-side comparison of top 3 cards. Comparison axes: Annual fee, Reward rate, Key benefits, Approval likelihood, Estimated annual savings. User can tap any card to go directly to that card's apply flow.

Step 8: Application Handoff

Pre-filled application using PAN, mobile, and name already collected. User completes remaining fields (address, employment details). Submitted to issuing bank application system via API.

Edge Case: If API handoff fails → save application state and prompt user to retry or continue via Bank website with a deep link.

10. Technical Architecture & Data Strategy

10.1 System Architecture

The architecture follows a layered pipeline pattern with strict data minimisation at each stage. All PII and financial signals are processed in isolated layers with role-based access controls.

- **Layer 1: Data Ingestion:** On-device SMS parser (iOS / Android native SDK); Bureau API connector (CIBIL / Experian); Optional AA framework connector (V2); linked transaction feed
- **Layer 2: Signal Extraction:** Structured event extraction (salary, merchant, category, amount, date); Feature engineering pipeline; Anonymisation and tokenisation before server transmission
- **Layer 3: Persona Engine:** Rule-based classification tree (V1); Input: behavioural feature vector; Output: {primary_persona, secondary_persona, confidence_score}
- **Layer 4: Recommendation Engine:** Weighted scoring model; Hard eligibility gate; Card catalogue API; Output: ranked card list with scores and explanations
- **Layer 5: Explainability Service:** Template-based NLG (Natural Language Generation) with dynamic variable injection; Maps model signals to user-facing copy
- **Layer 6: Presentation API:** RESTful API serving recommendation payload to the mobile app; Response includes: ranked cards, scores, explanations, pre-fill data

10.2 Data Flow & Privacy Architecture

- SMS messages are parsed entirely on-device using a local NLP / regex model. Only structured signals (e.g., 'salary_credit: ₹45,000, date: 2026-04-01') are transmitted, never raw message text.
- All data in transit is encrypted using TLS 1.3. Data at rest uses AES-256 encryption.
- User financial signals are stored with a 12-month retention window, then purged. Users can request deletion at any time via the app.
- Bureau data is fetched, used for scoring, and not persistently stored beyond the session unless the user proceeds to application.
- Role-based access controls (RBAC) ensure only the recommendation service can read financial signals. No human access to individual user signal data.

10.3 Compliance Framework

Regulation / Framework	How We Comply
RBI Digital Lending Guidelines	Consent collected before any data fetch. No algorithmic lending decision communicated as final recommendation is advisory, not a credit decision.
DPDP Act 2023	Explicit consent for each data category. Purpose limitation enforced. Right to erasure supported. Data fiduciary obligations met.
CIBIL / Experian API ToS	Soft pull only in V1. User informed before pull. Results not shared with third parties.
Account Aggregator (V2)	AA framework integration will require separate RBI-licensed AA entity agreement. Out of V1 scope.

11. Success Metrics & Targets

All metrics are tracked from Day 1 of launch. Targets are set for the 90-day post-launch window unless otherwise noted.

Metric	Baseline	V1 Target (90 days)	Cadence
BUSINESS METRICS			
Recommendation → Application CVR	~12% (industry)	25%	Monthly

Card Issuance Rate (applications approved)	~35% (industry)	50%	Monthly
Revenue per Approved Card (affiliate fee)	Baseline TBD	+20% vs baseline	Monthly
PRODUCT METRICS			
Onboarding Completion Rate	~40%	65%	Weekly
SMS Consent Acceptance Rate		70%	Weekly
Bureau Consent Acceptance Rate		60%	Weekly
Recommendation Page CTR		80% (of completions)	Weekly
AI / MODEL METRICS			
Persona Classification Accuracy		> 80% (post-hoc survey validation)	Monthly
Recommendation Acceptance Rate		70% (user selects recommended card)	Monthly
Approval Success Rate (recommended cards)		> 60%	Monthly
UX METRICS			
Time to Recommendation	~4 mins (current form)	< 90 seconds	Weekly
User Satisfaction Score (CSAT)		> 4.2 / 5.0	Monthly
Recommendation Trust Score (survey)		> 75%	Monthly

12. Risks & Mitigations

Risk	Likelihood	Mitigation
Low SMS consent rate	High	On-device parsing messaging; transparent consent UI; show value ('See how this helps') before consent ask; ensure graceful fallback so journey continues without SMS.
SMS data quality too sparse for reliable inference	Medium	Minimum signal threshold check before using SMS-derived features; fallback to explicit preference collection below threshold.
Bureau API latency or downtime	Medium	Async bureau fetch with a 5-second timeout; if failed, proceed without bureau signals and suppress eligibility probability display.

Recommendation accuracy below target	Medium	A/B test scoring weight configurations in first 30 days; instrument every recommendation-to-outcome for rapid feedback loop.
Regulatory change (DPDP Act enforcement)	Low–Medium	Modular consent architecture allows rapid updates to consent copy and data flows without core engine changes; legal review every quarter.
Credit card eligibility criteria changes breaking the engine	Low	Card catalogue stored as configuration, not hardcoded; a single config update propagates to all scoring and display logic.

13. Future Roadmap (Post V1)

V2

- ML-based persona clustering replacing rule-based engine (k-means / logistic regression on behavioural features).
- Account Aggregator (AA) integration for richer, real-time financial signals.
- Reward simulation tool: 'If you spend ₹X on this category, you earn ₹Y back per year'.
- Real-time recommendation refresh when financial signals change.

V3

- Conversational AI recommendation assistant (LLM-powered chatbot journey).
- Personalised financial coaching: 'Spend ₹2,000 more on dining this month to unlock lounge access'.
- Cross-sell into personal loans, FD-backed products, and insurance driven by the same behavioural engine.

V4

- Multi-bank and multi-issuer recommendation support.
- Proactive re-recommendation engine: notifies users when a better card becomes available based on evolving spend patterns.
- Portfolio optimisation: 'You have 3 cards here's how to use each one to maximise total rewards'.